Technology assessment

Comparing refining processes

Hydrometallurgy

- **Method**: Chemical leaching with acids
- **Input**: Black mass
- **Output**: Battery metals
- **Recovery rates**: 95% Ni and Co; 90% Li and Mn
- **Opex**: $1,600-2,300 per tonne of black mass
- Established at commercial scale
- Most profitable at higher volumes
- Most common recycling method

Pyrometallurgy

- **Method**: Blast furnace at 1,400°C
- **Input**: Batteries
- **Output**: Hydro needed to get Li and Mn from slag
- **Recovery rates**: 95% Ni and Co; ≥0% Li and Mn
- **Opex**: More expensive than hydro "Melt" of nickel, cobalt, copper alloy is intermediary
- Established at commercial scale
- Some examples but mostly in Europe and US

Electrochemical

- **Method**: Redox reaction with electricity
- **Input**: Black mass
- **Output**: Battery metals
- Proven in lab and pilot scale
- Can recover lithium
- Low opex, so viable for LFP recycling

Direct

- **Method**: Cathode-to-cathode recycling
- **Input**: Batteries/black mass
- **Output**: pCAM/CAM
- Proven at small scale

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Technology assessment

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Hydrometallurgy	Pyrometallurgy	Electrochemical	Direct
Li-Cycle	ACCUREC	NEU BATTERY MATERIALS	FARASIS
GEM 格林美 资源有限 循环无限 LIMITED RESOURCES UNLIMITED RECYCLING	GLENCORE	NTH CYCLE	OnTo-Technology Advancing Battery Recycling
성일하이텍(주) SungEel HiTech.	umicore		

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Technology assessment

Scale requirements across recycling methods

The minimum scale for smelting is higher than for mechanical-hydro metal extraction, which results in differences in the footprint of operations approaches but offers potential for smaller-scale operations handling specific chemistries.

	Metal extraction			Metallurgical refinement			
Recycling method	Equipment	Minimum industrial scale	Capex	Equipment	Minimum industrial scale	Capex	Footprint implications
Pyrometallurgy - hydrometallurgy	Smelter with additional off-gas processing	150,000 tpy of LiB	$200 mln	Dedicated Hydrometallurgical process	Typically driven by extraction capacity	$80 mln *(assuming 150,000 tpy of LiB)*	• Typically requires a centralized footprint, with batteries and/or black mass shipped to a central smelter, given the large minimum industrial scale • Hydrometallurgical refinement likely co-located with smelter to minimize logistics costs. Refinement capacity typically aligned with extraction capacity
Mechanical - hydrometallurgy	Shredding and separation line	60,000 tpy of LiB	$45 mln	Flexible hydrometallurgical process	Typically driven by extraction capacity	$170 mln *(assuming 60,000 tpy of LiB)*	• Metal extraction capacity typically driven by calcination process due to scale efficiency and permitting • Therefore, mechanical + hydrometallurgy processes with calcination stage typically require a centralized footprint for batteries to be shipped to for mechanical separation • Hydrometallurgical refinement likely co-located and capacity aligned with extraction capacity

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